

chief executive officer, Cygni Energy, says, "Solar-DC inverterless controller (named IBIS) promises forty to fifty per cent higher energy efficiency by incorporating inverterless technology at Solar 48V DC, a technology developed in collaboration with IITM. IBIS saves fifty per cent electricity bill through intelligent use of solar, battery and energy-efficient DC equipment and appliances."

Cygni mobile app allows customers to check their usage on-the-go. With Bluetooth/GPRS connectivity, they can track or monitor their energy consumption remotely through the dashboard.

Another solution, Dash solar controller for commercial applications, provides an energy-efficient power management and distribution system. Its smart control unit ensures maximum utilisation of solar power and supplements it with power from the grid, besides providing a battery backup.

For catering to both high- and low-

Technical benefits

- By using 48V DC solution, we can avoid power conversions.
- No carbon foot prints; 100 per cent green energy.
- 48V DC falls under safe extra low voltage (SELV); thus no shock, reducing risk of electric shock and no fire accidents.
- Controlling/monitoring of loads with IR remote controls.
- Monitoring power savings and readings in energy meter.
- High efficiency.

power applications, Rajaraman says, "Typically, homes have two distinct lines: 5A and 15A. Our vision is to convert 5A line into a DC line and keep high-power appliances like dishwashers, air-conditioners and the like, which are typically on 15A line, intact."

Besides these, other products include AC-DC converters and compatible 48V DC appliances. The converters come in three variants, for 2.4kW, 500W and 150W power output. With 48V DC appliances like BLDC fans, LED

lights and bulbs, and LED TVs, existing AC wiring can be used.

About pricing, Rajaraman says, "Price of our products ranges from ₹ 20,000 to ₹ 100,000, and changes based on the bundle of appliances that we provide to customers."

Discussing the challenges, Rajaraman, says, "Our customers are majorly in hard-to-reach rural areas. To handle the challenge of post-installation service, we have a dedicated onsite and online support team for quick serving, and all products come with built-in GSM/GPRS, which enables remote monitoring even in such rural areas."

The company was the first startup to benefit from the government's Startup India Action Plan when it was launched. Talking about the future, Rajaraman says, "The rural productive appliances segment is moving towards DC in a big way. Our major focus is on finding more domestic and productive appliances that run effectively on our system."

—Ayushee Sharma

4 ROBOTS AS A COMMUNICATION INTERFACE

Our everyday technology can take inputs from the environment to decode the meaning of our behaviour and turn information into digital cues for devices to respond accordingly. This makes interactions between humans and machines so natural and seamless that communicating with machines feels like talking to humans or live animals. Now that we live in a more connected society, it is becoming easier to make robots talk to various sensors, mobiles and computers in our living spaces to understand user behaviour and support users.

While robots are playing an increasingly important role in manufacturing, deliveries, disaster responses, security and surveillance and many other fields, Yukai Engineering Inc. is focused on building the consumer market for personal robots that can enhance the quality of daily life. Its latest innovations are BOCCO and QOOBO.

BOCCO cloud API makes it easy to connect to Internet of Things (IoT) sensors and control the robot

remotely. Various companies are using BOCCO to deliver different services and information, including monitoring the elderly, kids and pets, smart home appliances control, weather forecasting and emergency alerts.

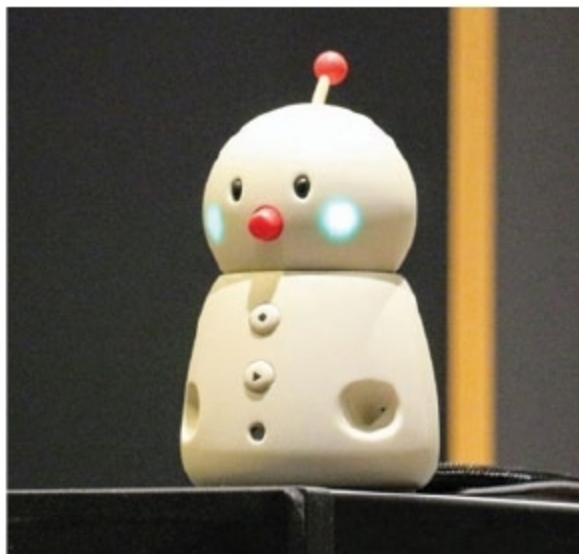
It has an easy-to-use voice-enabled user interface with effective voice commands, and the robot's responses (movements, LED lights blinking patterns, etc) can be clearly understood by the user. It uses voice processing

technology that can effectively deal with noise and other problems.

BOCCO watches over kids and elders at home when you cannot be there. BOCCO app makes it easy to exchange messages with them. Messages sent are read out by BOCCO in a cute loud voice for them. What they say on BOCCO in reply to your messages is sent to your phone. The vibration sensor that comes with BOCCO notifies you to receive your child when he or she comes home, and opens the door. Its price is around ₹ 16,000.

The next-generation communication robot BOCCO emo, which inherited the name BOCCO, has changed to a round, sitting-down appearance. The range of expressions has further increased, such as LED on the cheek blinking and head antenna falling. The robot understands the language spoken by the user and can speak in its robot language, BOCCO.

The artificial intelligence (AI) conversation engine, which is the basis for BOCCO language, has been



BOCCO (Credit: www.digimonostation.jp)